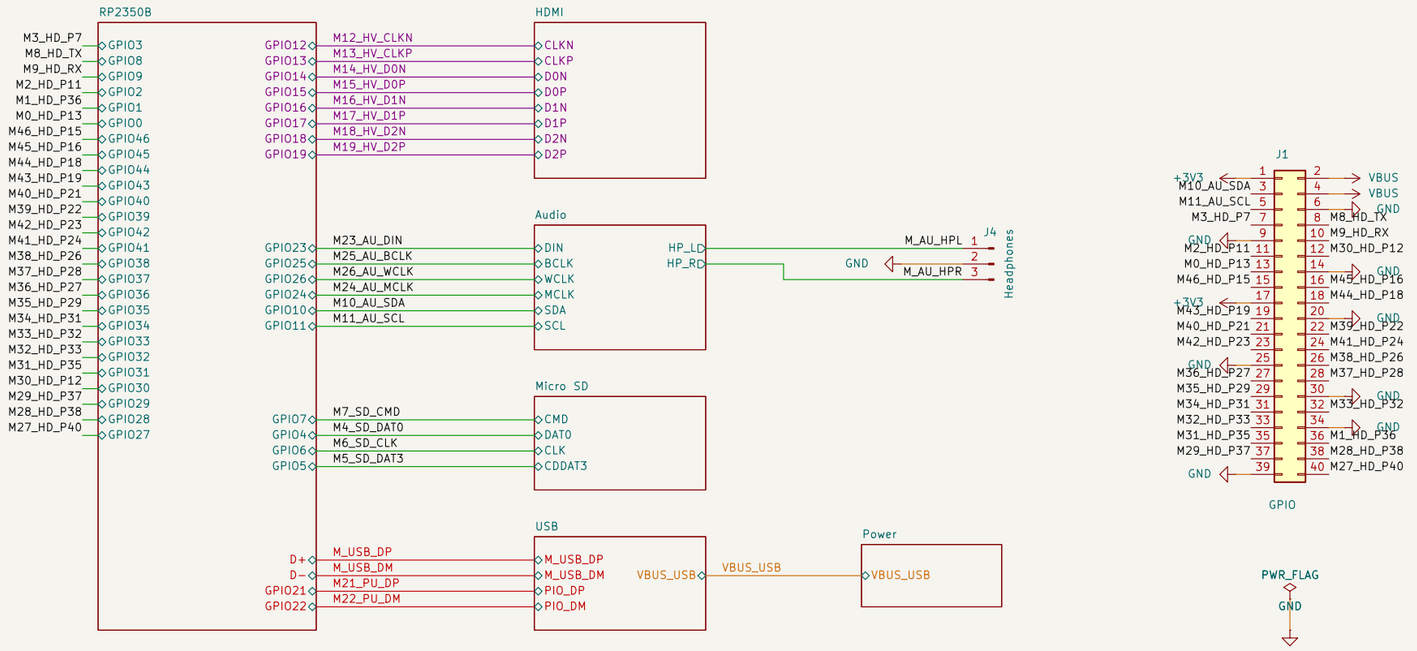


J1: Raspberry Pi 40-pin layout, pins mapped by function to RP2350B peripherals (net names give the J1 pin):
I2C1 3/5 (GPIO10/11, shared with the codec), UART1 8/10 (GPIO8/9), SPI as PI SPI0 19/21/23/24 = SPI1 TX/RX/SCK/CSn (GPIO43/40/42/41), CE1 26 (GPIO38).
PCM/I2S 12/35/38/40: BCLK GPIO30, LRCLK GPIO31 (consecutive, for PIO I2S), DIN GPIO28, DOUT GPIO27.
HAT ID EEPROM 27/28 = I2C0 SDA/SCL (GPIO36/37). 5V pins = VBUS rail, 3V3 pins = +3V3.

Block codes (sheet title blocks carry the same code):

M	RP2350B core, flash, crystal, buttons		(sheet RP2350B)
PR	PSRAM U1 (CS = GPIO47)	LD	LEDs: LD2 blue GPIO20 (status), LD1 green (power)
HV	HDMI (mini-HDMI J6)	SD	micro SD J5 (SPI mode)
AU	TLV320DAC3100 codec, speaker J3 (JST-GH), headphones J4 (L/GND/R)		
UC	native USB-C J8 (RP2350 USB, 5 V in)	PU	PIO-USB host USB-C J7
PW	power: polyfuse, Schottky, SY8089 3V3 buck, test points		
HD	40-pin header J1: UART, SPI, I2C, GPIO (net names give the J1 pin); SWD on J2 (JST-SH, sheet RP2350B)		

GPIO map:

GPIO0-3	header 13/36/11/7	GPIO4-7	micro SD (SPI0)
GPIO8/9	UART1 TX/RX (header 8/10)	GPIO10/11	I2C1 SDA/SCL (header 3/5, codec)
GPIO12-19	HDMI TMDS (HSTX)	GPIO20	LD2 status LED
GPIO21/22	PIO-USB D+/D- (J7)	GPIO23-26	codec DIN/MCLK/BCLK/WCLK
GPIO27-46	header (SPI1 40-43 on 21/24/23/19, CE1 38 on 26, HAT ID 36/37 on 27/28)		
GPIO47	PSRAM CS	SWD on J2 (JST-SH)	HP L / GND / R on J4

H1 H2 H4 H3

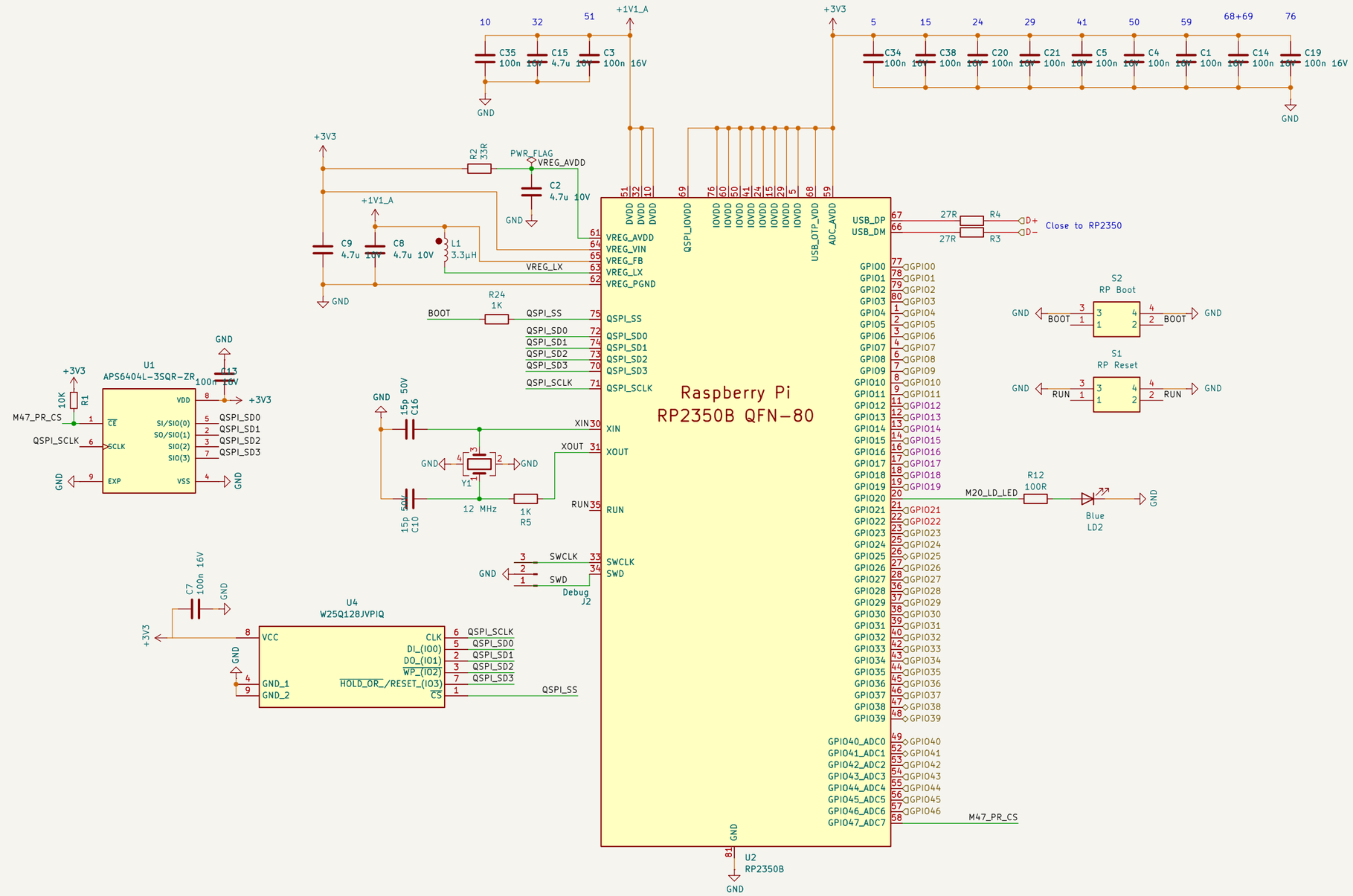
Net names: <M><gpio>_<block>_<signal> (M = RP2350B); block codes in the legend
https://github.com/rh1tech/frank

Mikhail Matveev

Sheet: /
File: frank_z0.kicad_sch

Title: FRANK Z0

Size: A3	Date: 2026-11-09	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 1/7



Block code: M (RP2350B core), PR (PSRAM, CS = GPIO47), LD (LED GPIO20)
https://github.com/rh1tech/frank

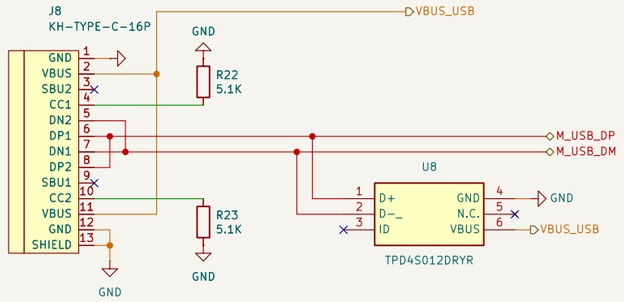
Mikhail Matveev

Sheet: /RP2350B/
File: rp2350b.kicad_sch

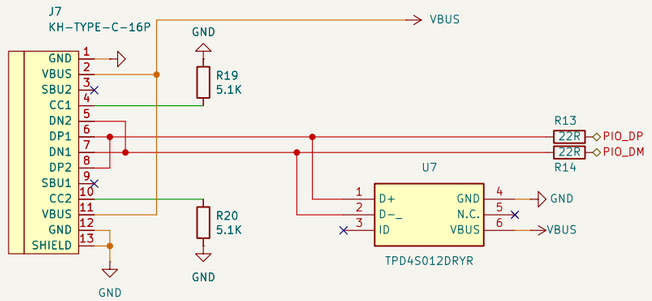
Title: FRANK Z0

Size: A3	Date: 2026-11-09	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 2/7

J8 native USB-C (RP2350 USB device / BOOTSEL, 5 V power in): CC 5.1K = sink, U8 TPD4S012 ESD at the connector.
Series 27R for USB_DP/DM sit at the RP2350 (MCU sheet).

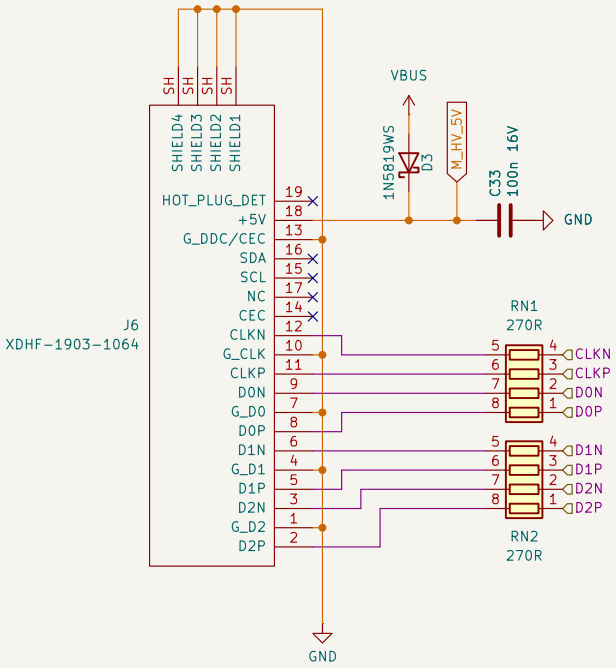


J7 PIO-USB host (as on Waveshare RP2350-PiZero): CC 5.1K, U7 TPD4S012 ESD, R13/R14 22R series.
GPIO21 = D+, GPIO22 = D-. VBUS comes from the board VBUS rail (5 V to the device).



Block code: UC (native USB-C), PU (PIO-USB)
https://github.com/rh1tech/frank
Mikhail Matveev

Sheet: /USB/		
File: usb.kicad_sch		
Title: FRANK Z0		
Size: A3	Date: 2026-11-09	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 3/7



J6: mini-HDMI (Type C, Lian Xin XDHF-1903-1064, same land pattern as XUNPU HDMI-201); symbol pin numbers follow the Type C pinout.
RN1/RN2 270R TMDS series as on FRANK Mini; HPD/DDC/CEC not used.
HDMI +5V (pin 18, net M_HV_5V): diode-OR of VBUS through D3 and VBUS_USB_F through D2 (both 1N5819WS),
so it skips D1 on J8 power (~4.7 V); a powered sink cannot back-feed the board.

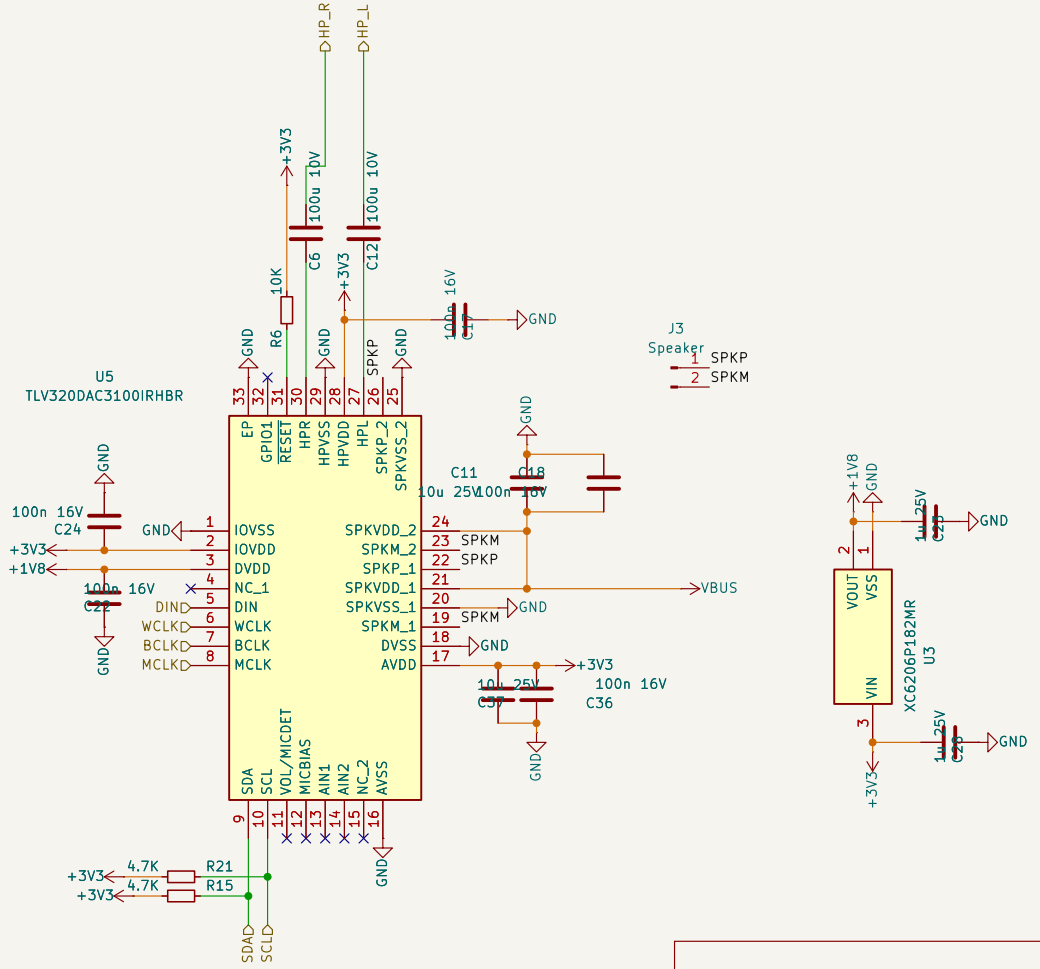
Block code: HV (HDMI)
https://github.com/rh1tech/frank
Mikhail Matveev

Sheet: /HDMI/
File: hdmi.kicad_sch

Title: FRANK Z0

Size: A4	Date: 2026-11-09	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 4/7

U5 TLV320DAC3100 (from FRANK Next): I2S DIN/BCLK/WCLK + MCLK, I2C control (4.7K pull-ups), RESET via 10K pull-up.
Class-D speaker out -> J3 JST-GH (SPKP/SPKM, 8R speaker).
Headphone out HPL/HPR -> C6/C12 100u DC blocking -> M_AU_HPL / M_AU_HPR, out to header J4 (L / GND / R).
SPKVDD straight from VBUS; DVDD 1.8 V from the XC6206 LDO.



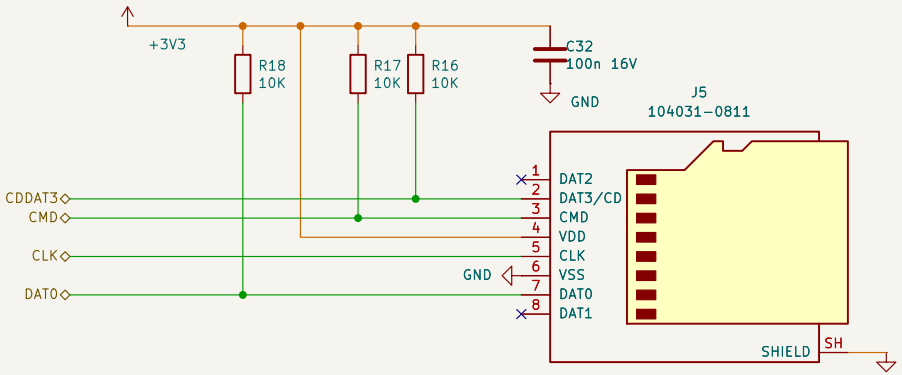
Block code: AU (audio codec)
https://github.com/rh1tech/frank
Mikhail Matveev

Sheet: /Audio/
File: audio.kicad_sch

Title: FRANK Z0

Size: A4 | Date: 2026-11-09 | **Rev: 1.00**
KiCad E.D.A. 10.0.1 | Id: 5/7

10K pull-ups on CS, CMD, DAT0 (SPI); DAT1/DAT2 unused; C32 decouples the socket VDD

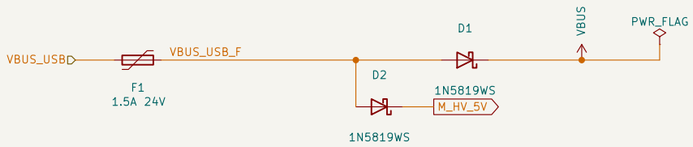


Block code: SD (micro SD, SPI mode)
https://github.com/rh1tech/frank
Mikhail Matveev

Sheet: /Micro SD/
File: sd.kicad_sch

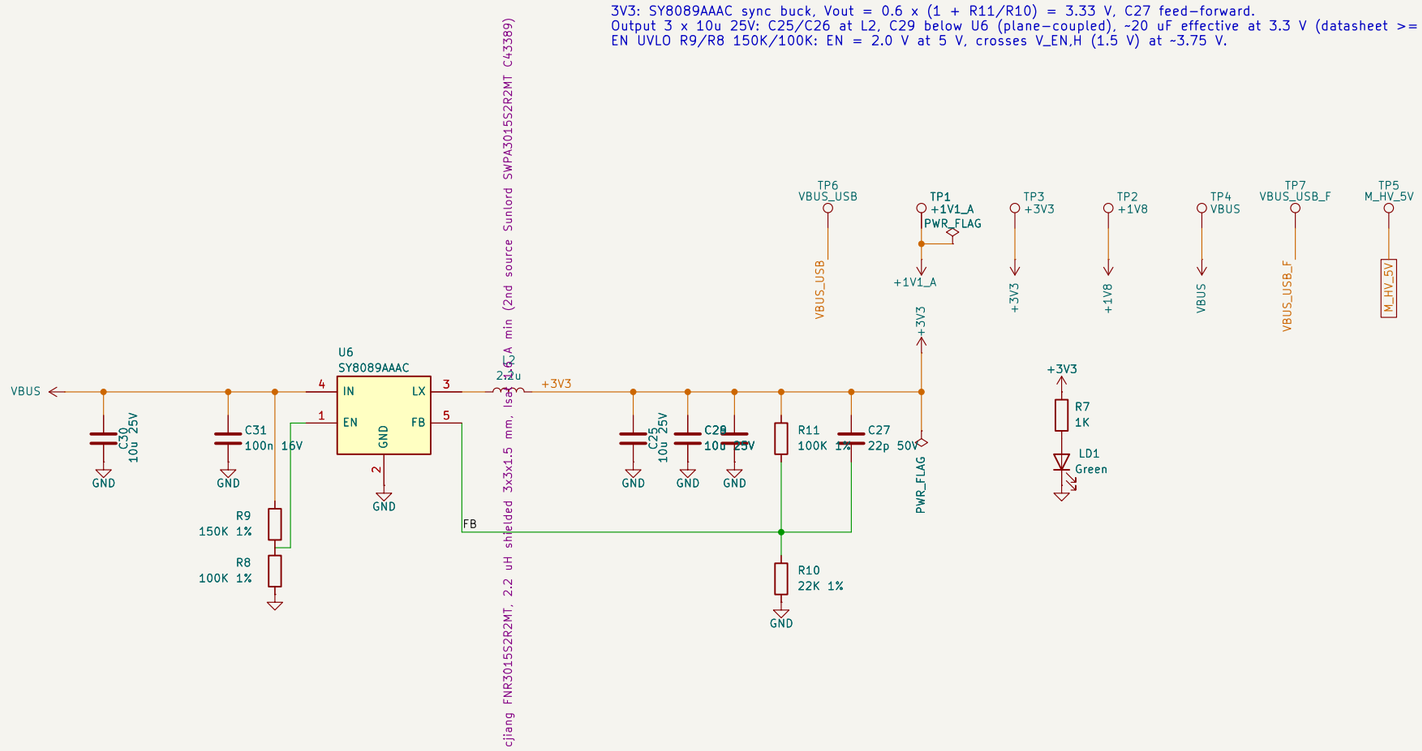
Title: FRANK Z0

Size: A4	Date: 2026-11-09	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 6/7



5 V in: J8 (native USB-C) -> VBUS_USB -> F1 polyfuse -> D1 Schottky -> VBUS rail.
D1 blocks back-feed into J8 when the board is powered from J7 (PIO-USB).
VBUS rail feeds J7 VBUS (5 V to the USB device), HDMI +5V (D3), audio SPKVDD and the 3V3 buck (C30/C31 at its input).
D2 feeds HDMI +5V (M_HV_5V) straight from VBUS_USB_F, one diode drop instead of two.

3V3: SY8089AAAC sync buck, Vout = 0.6 x (1 + R11/R10) = 3.33 V, C27 feed-forward.
Output 3 x 10u 25V: C25/C26 at L2, C29 below U6 (plane-coupled), ~20 uF effective at 3.3 V (datasheet >= 22 uF).
EN UVLO R9/R8 150K/100K: EN = 2.0 V at 5 V, crosses V_EN,H (1.5 V) at ~3.75 V.



Block code: PW (power)
https://github.com/rh1tech/frank
Mikhail Matveev
Sheet: /Power/
File: power.kicad_sch

Title: FRANK Z0

Size: A3	Date: 2026-11-09	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 7/7